

September AI Engineering Sprint

August 12–31 you became a builder. September 1–30 you become a shipper.
A focused 4-week program to build reliable, cheap, and used AI products.

Core Path: Evaluation & Observability → Advanced Agents & Workflows → Multimodal / Voice / Real-time
→ One Real Product with Users
Time: 3-4 hrs/day | **Goal:** One live product + cost/latency dashboard + advanced agent repo

WEEK 1 — Sept 1–7: Make Systems Reliable and Cheap

What separates tutorials from senior engineers: evals, traces, and cost control.

Sept 1–2: Deep Evaluation & Observability

- Instrument your August RAG + Agent with Langfuse or LangSmith: trace every LLM call, tool call, latency, cost
- Build eval set of 50 questions. Calculate Precision@k, Faithfulness, Answer Relevancy (RAGAS or manual)
- Add cost dashboard: tokens/query, \$/1k queries

Deliverable: /evals folder with traces + cost report

Sept 3–4: Latency & Cost Engineering

- Semantic caching with Redis for repeated queries
- Query routing: small model (Haiku / 4o-mini) for simple, large model for complex
- Parallel tool execution + streaming

Deliverable: Same app, 40% cheaper and 2x faster. Document it.

Sept 5–7: Data Flywheel

- Ingestion pipeline: Watch folder / S3 / Notion → chunk → embed → pgvector with versioning
- Feedback loop: thumbs up/down → DB → nightly eval report
- Study fine-tuning vs RAG vs prompt — when to use what

Deliverable: Blog post / README: 'Why I didn't fine-tune'

WEEK 2 — Sept 8–14: Advanced Agents

Sept 8–10: Workflow Orchestration

Stop single-agent loops. Build a real workflow with LangGraph or Node.js state machine.

- Option A - Deep Research Agent: Planner → Researcher (parallel web search) → Critic → Writer
- Option B - Coding Agent: Planner → Coder → Tester → Fixer

Deliverable: GitHub Repo #5 - Multi-step agent with visual graph

Sept 11–12: Human-in-the-loop + Long-running

- Add checkpoints, approval steps, and resume capability
- Add background jobs with BullMQ / Inngest. Agent runs 10 mins, user leaves

Deliverable: Agent that emails you when done

Sept 13–14: MCP in Production

- Build 2 useful MCP servers: 1) Your Postgres + RAG 2) GitHub / Gmail / Calendar
- Connect to Claude Desktop / Cursor. Make coding agent use your own data

Deliverable: Published MCP server with npx install

WEEK 3 — Sept 15–21: The Edge — Voice, Realtime, Multimodal

Sept 15–17: Voice + Realtime

- Stack: STT (Deepgram/Whisper) → LLM → TTS (ElevenLabs) + interruption handling
- Use LiveKit or Vapi or WebRTC + Node.js

Deliverable: Talk to your RAG documents

Sept 18–19: Multimodal

- Accept images/PDFs: screenshot → GPT-4o / Gemini → extract + answer
- Add image generation tool to your agent

Deliverable: Chat app accepts images, PDFs, voice

Sept 20–21: Memory

- Implement: short-term (chat history) + long-term (vector store of user facts) + entity memory

Deliverable: Agent remembers you across sessions

WEEK 4 — Sept 22–30: Ship ONE Real Product

No more demo repos. One product people can actually use.

Sept 22–24: Pick ONE and Commit

Don't build 3. Pick ONE:

- AI Interviewer for full-stack devs
- RAG over Bangalore startup docs for founders
- AI code review bot for GitHub PRs
- Voice customer support for a local business

Architecture: Next.js → Auth (Clerk) → Stripe → Your Agent/RAG → Postgres + pgvector → Fly.io / Railway + Vercel

Sept 25–27: Ship It

- Docker, CI/CD, rate limiting, auth, billing, analytics, Sentry. Real README, real landing page
- Launch on Twitter, LinkedIn, Product Hunt or send to 10 real users

Deliverable: Live URL

Sept 28–29: Content + Proof

- Record 3 short demos (60-90 sec each)
- Write one deep technical post: 'How I built X and how I eval it'

Sept 30: The Hiring Exam

Be ready to answer:

- How much does 1000 queries cost and how did you reduce it?
- How do you know if your RAG is degrading in prod?
- What happens if a tool returns PII or malicious instructions?
- Why LangGraph over simple loop for this use case?
- Show a trace of your agent failing and recovering

Success Criteria & Guardrails

By Sept 30, you must have:

- One product live with auth + payments + 10 real users
- Cost / latency / trace dashboards (Langfuse / LangSmith)
- One advanced multi-step graph-based agent
- One voice or vision feature shipped
- One strong technical write-up explaining tradeoffs

September Do-Not-Touch List

- Learning 10 AI frameworks at once
- Starting 3 new projects and finishing 0
- Building without evals
- Adding multi-agent when single agent works
- Rebuilding UI for the 10th time
- Watching 'AI will replace engineers' content
- Collecting certificates instead of shipping

The Core Skill Stack — September

LLM → Eval → Observability → Cost Optimization → LangGraph → Human-in-loop → MCP Prod → Voice/Realtime → Memory → Shipping + Monetization

Built for Kai Karthik — August Sprint Graduate. From builder to shipper. September is about users, not stars.